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Subject: Formulas and Abbreviations on Siemens Electronic Trip Units

While QT/EPEC generally does not configure breaker settings at the factory and cannot advise on settings specific to individual projects or applications, we have prepared this brief guide to assist in navigating the abbreviations, formulas, and nomenclature on Siemens electronic trip units utilized in our products.

Trip Unit Functions:

L - Long-term overcurrent protection

S - Short-term overcurrent protection

I - Instantaneous overcurrent protection

G - Ground fault protection (defaults to vector summation unless an external CT is specified)

N - Neutral protection (requires 5xx or 8xx series ETU and external CT on 3 pole breakers)

Functional Abbreviations:

I_n - Frame rating of breaker, not to be confused with the trip rating.

I_r - Long-term overload (not fault) setting.

t_r - Long-term overload delay.

I_{sd} - Short-term overload/minor fault setting.

t_{sd} - Short-term time delay.

I_i - Immediate (hard fault) setting.

t_{id} - Immediate time delay (see notes).

I_N - Neutral overload protection.

Breaker Setting Types:

I_r /A - Long-term overload setting in amps.

I_r = x I_n - Long-term overload setting as a factor of the frame rating. For example, if I_n = 4000, the .5 position is 4000A x .5, or 2000A.

t_r /s - Long-term overload delay time in seconds.

t_r @ 6x I_r /s - Overload delay time in seconds *at six times the long-term overload setting*. This is an *inverse curve* setting that varies long-term overload time in relation to overload value. For example, if I_n = 4000A, I_r = .5 x I_n (2000A), and this switch is set to the “8” position, then the delay time is 8 seconds at 12,000A. The delay time will increase as the amperage is lower, and decrease as the amperage gets higher, until it reaches the short-term threshold.

t_R(s) (@6x I_R) - ETU745 legacy parameter. Same concept as the previous one, but uses a switch above the setting to toggle between I²t and I⁴t. The settings on the left side of the dial apply to I²t and the settings on the right side of the dial apply to I⁴t.

I_{sd} = x I_r - Short-term overload setting as a factor of long-term trip setting. For example, if I_r = 1000, the 5 position is 1000A x 5 or 5000A.

I_{sd} = x I_n - Short-term overload setting as a factor of frame rating. For example, if I_n = 4000, the 6 position is 4000A x 6 or 24,000A.

t_{sd} / s - Short-term overload delay time in seconds.

I_i / A - Immediate fault setting in amps.

$I_i = x I_n$ - Immediate fault setting as a factor of the frame rating. For example, if $I_n = 4000$, the 6 position is 4000A x 6, or 24,000A.

$I^2 t_{sd} / s$ - Short-term overload delay time in seconds, with two functions. The right, or FIXED side of the rotary switch are fixed time increments in fractions of a second. The left side of the dial sets the median of an inverse curve based on fault current. Fault current higher than I^2 will result in a time shorter than the setting, and fault current lower than I^2 will result in a time longer than the setting.

$t_{sd}(s) I^2 t_{sd}(s)$ - ETU745 legacy parameter. Same concept as above but operates in reverse. The left side of the dial is FIXED values, and the right side applies an I^2 curve at $12x I_n$.

$I_N = x I_r$ - Neutral trip as a factor of long-term trip setting. For example, if $I_r = 1000$, the 1.25 position is 1000 x 1.25, or 1250A. This setting may be present on all large frame 3VA LSIg trip units, but **only functions** on the following 3VA breaker configurations:

- 3 pole with ETU5xx or ETU8xx **and** the appropriate external neutral CT.
- 4 pole with all ETU series.

I_N (ETU745 legacy slide switches) - The first switch enables or disables neutral protection. The second switch sets neutral protection at either .5x or 1x the option plug rating.

MEMORY (ETU745 legacy slide switch) - This switch enables or disables the thermal memory function of the trip unit. Thermal memory protects the breaker from assuming a full cool down when there is a short duration between trip events.

Notes:

The rotary switches on ETU300 and ETU600 trip units for 3WA breakers function as an override of the electronic settings. If the operator wishes to use settings programmed through the ETU display or PowerConfig software, the **e.SET** position must be used on any applicable switches.

Rotary switches on all Siemens electronic trip units *must* be set between, and not on, the lines. Setting a rotary switch on a line will result in the trip unit potentially "jittering" between two settings, potentially causing nuisance tripping or other undesirable results.

t_{id} is *rarely* used on UL listed equipment and is generally only seen on components shared between both UL and IEC listed versions of the product.

Questions? Reach out to support@epecsolutionsinc.com for more information!